

ANEF: Adversarial Neural Encryption Framework for Secured Consumer Electronics in Smart Cities

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Abstract—Secure communication in resource-constrained Internet of Things (IoT) deployments—such as consumer electronics and smart city infrastructures—faces persistent challenges from both limited processing capacity and increasingly adaptive cyber threats. This work introduces the Adversarial Neural Encryption Framework (ANEF), which models encryption as a non-invertible mapping trained through an adversarial three-agent setup comprising an encoder (Alice), a decoder (Bob), and an adversary (Eve). The framework integrates entropy-regularized encryption, dynamic key scheduling with periodic reseeding, and quantization-aware optimization to support 8-bit inference on constrained hardware. Adversarial resistance is reinforced through residual masking and curvature-based penalties, improving robustness against adaptive attacks. Training is guided by a hybrid objective that balances reconstruction fidelity, orthogonality preservation, and adversarial minimization, applied over sequential encryption states. Experiments using the UNSW-NB15 dataset evaluate accuracy, communication overhead, and adversary success rate across diverse packet sizes. ANEF achieves 94.7% decryption accuracy with a 10.3% transmission overhead, while maintaining a 4.1% adversary success rate.

Index Terms—Secure transmission, Internet of Things, consumer electronics, cyber threats, trustworthy, smart cities.

I. INTRODUCTION

THE large-scale increase of Internet of Things (IoT) nodes in smart cities and consumer electronics has introduced significant heterogeneity in communication protocols, sensor configurations, and operational constraints across urban environments [1]. This shift toward pervasive sensing and real-time actuation has elevated the sensitivity of transmitted data, demanding robust and reliable security

primitives across the communication stack [2]. However, ensuring cryptographic confidentiality and data integrity across constrained, intermittently connected, and low-power IoT devices remains a significant engineering challenge in the face of emerging cyber threats [3].

A. Related Work

Studies addressing security in heterogeneous IoT environments explore classical encryption, federated learning, and privacy-preserving computation. Attribute-based encryption with user revocation enables fine-grained access in cloud-assisted IoT [4], while fully homomorphic encryption supports secure federated learning [5]. Elliptic curve-based key exchange with logistic regression improves integrity validation [6]. Network-layer perturbation obfuscates message semantics [7], and blockchain–SDN hybrids detect volumetric attacks using neural classifiers [8]. Graph neural networks over blockchain further enhance authenticated transaction control [9]. Resource coordination models emphasize scalability and adaptive security in IoT communication. Lightweight authentication and key agreement schemes address latency-sensitive industrial scenarios [10], while hierarchical access control supports constrained home automation devices [11]. Federated learning with quantum-safe primitives enables dynamic resource allocation [12]. Energy-aware routing with aerial relays improves secrecy capacity [13], and data aggregation frameworks integrating dissemination control and real-time validation enhance secure wireless sensor operations in dense networks [14]. Table I provides a comparative overview of key capabilities addressed across existing and proposed approaches.

B. Research Motivation & Problem Definition

This study addresses the challenge of securing end-to-end data transmission in resource-constrained IoT environments, particularly within consumer electronics–dense smart cities operating under dynamic, high-density deployment conditions [15]. While prior research has examined intrusion detection, anomaly classification, and lightweight cryptography as separate problems, the integrated question of autonomous, data-driven, and reliable security generation remains insufficiently explored [16]. The motivation originates from the ongoing trade-off between achieving secure transmission and meeting the operational constraints of smart-city IoT and consumer electronics deployments [17]. These devices routinely exchange sensitive data such as mobility traces,

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TABLE I
COMPARATIVE ANALYSIS OF EXISTING APPROACHES
AND THE PROPOSED ANEF FRAMEWORK

Ref.	Identified Limitation	Resolution in Proposed ANEF Framework
[5]	High computational cost and slow inference on constrained devices	Uses lightweight neural inference with quantization-aware training for efficient deployment
[4]	Static policy bindings and key overhead in dynamic device populations	Implicit encoding via learned representations without policy-dependence
[6]	Statistical model offers fixed transformation and no adversarial adaptability	Adversarial training induces dynamic transformation resilient to inference attempts
[7]	Perturbations can degrade legitimate signal accuracy or violate protocol conformity	Decoy packets retain statistical structure while preserving legitimate message fidelity
[9]	Requires persistent global state synchronization; high latency	Localized neural transformation avoids synchronization dependencies
[10], [11]	Dependence on discrete key exchange, vulnerable to session hijacking	Keyless contextual encoding using shared initial weights, reducing exchange vulnerabilities
[13]	Infrastructure-dependent and context-specific optimization; not generalizable	Framework remains topology-agnostic and communication-model independent
[8]	IDS modules are reactive and detection-dependent	Encoding actively disrupts inference before detection is required

environmental metrics, and infrastructure states. Unauthorized interception or alteration of this information can undermine service reliability and compromise operational safety under adversarial conditions [18].

C. Overview of the ANEF Framework

The framework introduces an encryption model based on adversarial neural optimization, comprising three jointly trained networks under a minimax formulation. The encoder applies a learned nonlinear transformation to sensor data, initialized from a shared seed. A symmetric decoder reconstructs the original message, while an adversarial network approximates unauthorized decryption attempts. Training maximizes the performance gap between legitimate and adversarial reconstruction to preserve communication integrity. To reduce passive inference risk, decoy packets statistically consistent with ciphertext are inserted into transmissions. The system supports quantized, low-latency inference for constrained IoT and consumer electronics devices, operating without iterative decoding or continuous key exchange to maintain confidentiality and computational efficiency in operational deployments. The main contributions of the proposed approach are summarized as below:

- 1) An adversarial encryption–decryption framework in which the cryptographic mapping is learned from data rather than defined by fixed rule.
- 2) A generative module that produces realistic IoT traffic patterns to accompany encrypted messages.
- 3) A deployment-ready pipeline capable of performing encryption and decryption in a single forward pass with low latency, designed for low-power IoT and consumer electronics hardware.

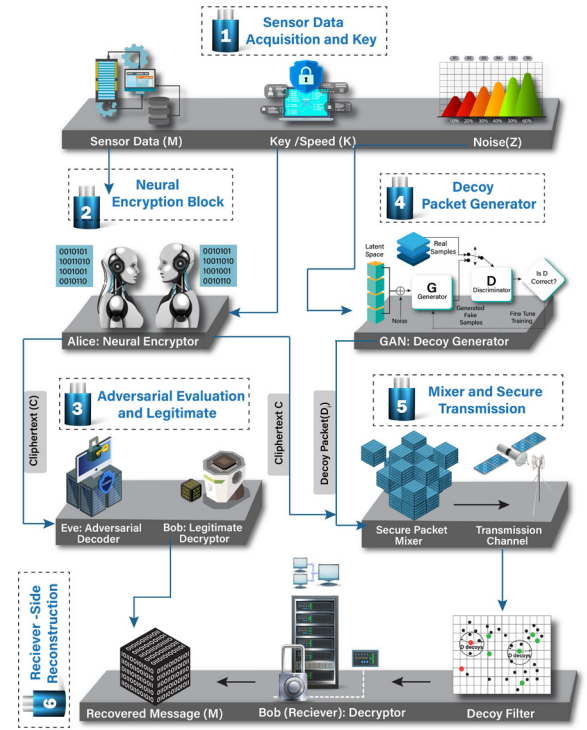


Fig. 1. System-level architecture of the proposed Adversarial Neural Encryption Framework (ANEF).

D. Structure of the Article

The structure of this article can be summarised as follows. Section I-A reviews existing literature on secure data transmission and neural cryptography within IoT systems. Section II details the proposed ANEF, including its architectural components and operational principles. Section III presents the simulation setup and performance evaluation. Section IV provides an ablation-based investigation. Finally, Section V concludes the study.

II. PROPOSED APPROACH: ADVERSARIAL NEURAL ENCRYPTION FRAMEWORK (ANEF)

The approach starts with a co-trained neural encryption network that encodes plaintext sensor input via a nonlinear transformation guided by a shared initialization key. The transformation is learned adversarially through competition between the encoder and a decoding network that simulates an unauthorized interceptor. In parallel, a decoder is trained to reconstruct the original signal with high fidelity, enabling reliable decryption for legitimate receivers in consumer-electronics-rich IoT environments. The adversarial constraint keeps the encoded representation intelligible only to the authorized decoder while opaque to external inference. As shown in Figure 1, all components are quantized post-training to ensure inference efficiency on constrained consumer electronics.

Following encryption, the system introduces synthetic decoy packets that statistically mimic genuine ciphertext sequences but carry no semantic payload. These decoys, generated by a secondary GAN unit, are randomly inserted into the outgoing data stream to mitigate passive adversaries. The receiver

applies selective filtering using synchronized sequence cues and reconstructs only valid encrypted content.

A. Neural Encryption via Adversarial Training

This module reformulates encryption as a non-invertible transformation constrained by adversarial dynamics in a variational setting, where the encoder Φ_α maps input message M into high-entropy ciphertext C while optimizing separation from any possible inverse mapping under Eve's decoding model D_β . A dynamically conditioned key schedule κ introduces structured noise synchronization without relying on external entropy injection, while a hybrid multi-objective loss embeds resilience across both spatial and temporal channels.

$$C = \Phi_\alpha(M, \kappa) = \tanh(\mathbf{W}_e \cdot \sigma(\mathbf{A}M + \kappa) + \mathbf{b}_e) \quad (1a)$$

$$\kappa = \mathbf{R}_\omega \cdot \text{ReLU}(\sin(\gamma t + \theta)) \quad (1b)$$

Equation 1a defines the neural encryption map Φ_α , where message $M \in \mathbb{R}^n$ and initialization key $\kappa \in \mathbb{R}^d$ are nonlinearly transformed using trainable weights $\mathbf{W}_e$, $\mathbf{A}$, and bias $\mathbf{b}_e$. Equation 1b specifies a dynamic key scheduling rule with sinusoidal modulation and learned vector $\mathbf{R}_\omega$ to encode temporal variation.

$$\mathcal{L}_{\text{adv}} = \min_{\alpha} \max_{\beta} \mathbb{E}_M \left[\|M - \Psi_\beta(C)\|_2^2 - \eta \cdot \log(1 - D_\beta(C)) \right] \quad (2a)$$

$$\mathbf{A} = \arg \min_{\mathbf{A}} \left(\det(\mathbf{I} + \mathbf{A}^\top \mathbf{A}) - \lambda \cdot \text{Tr}(\mathbf{A}^\top \mathbf{A}) \right) \quad (2b)$$

$$D_\beta(x) = \sigma(\mathbf{W}_d \cdot \text{ReLU}(\mathbf{U}x + \mathbf{b}_d)) \quad (2c)$$

The adversarial loss in Equation 2a optimizes encoder-decoder cooperation against a decoding adversary, balancing fidelity and attack resistance via η . Equations 2b–2c regulate the encoder's structure through determinant-trace minimization and define the discriminator's ReLU-based transformation for attack simulation.

$$\zeta = \nabla_\alpha \mathcal{L}_{\text{adv}} + \rho \cdot \|\nabla_\alpha \Phi_\alpha(M, \kappa)\|_F^2 \quad (3a)$$

$$\mathcal{E}_{\text{sec}} = \mathbb{E}_{M, \kappa} \left[\exp\left(-\|M - \Psi_\beta(C)\|_2^2\right) \right] \quad (3b)$$

$$\mathcal{Z}_\Phi = \int_{\mathcal{D}} \left| \frac{\partial^2 \Phi_\alpha}{\partial M^2} \right|^2 dM \quad (3c)$$

Equations 3a–3c control the gradient behavior of the encoder, ensuring smoothness and nonlinearity via second-order constraints and security-focused expected decryption failure.

$$\Upsilon(M, \kappa) = \text{mod}(\text{Hash}(M \oplus \kappa), \pi) \cdot \mathbf{v} \quad (4a)$$

$$\mathcal{J}_{\text{disj}} = \mathbb{E}[\|\Phi(M_1, \kappa) - \Phi(M_2, \kappa)\|_2^2 : M_1 \neq M_2] \quad (4b)$$

$$\mathcal{F}_\lambda = \|\text{FFT}(\Phi(M, \kappa))\|_2^2 - \lambda_{\text{fft}} \quad (4c)$$

$$\mathcal{B}_{\text{entropy}} = - \sum_j \mathcal{C}_j \log(\mathcal{C}_j + \epsilon) \quad (4d)$$

Equations 4a–4d ensure output diversity, frequency robustness, and entropy conservation. Υ tracks content-dependent signatures, $\mathcal{J}_{\text{disj}}$ promotes ciphertext separation, and $\mathcal{F}_\lambda$

uses frequency-domain properties to prevent structural predictability.

$$\Theta_{\text{enc}} = \arg \min_{\alpha} [\mathcal{L}_{\text{adv}} + \mu_1 \mathcal{Z}_\Phi + \mu_2 \mathcal{E}_{\text{sec}} + \mu_3 \mathcal{F}_\lambda] \quad (5a)$$

$$\mathcal{L}_{\text{info}} = \|\nabla_M \Phi_\alpha(M, \kappa)\|_1 - \log(1 + \mathcal{J}_{\text{disj}}) \quad (5b)$$

Final training uses a composite objective in Equation 5a, combining adversarial loss, second-derivative penalty, security bounds, and frequency constraints. Equation 5b encourages sparse gradients and diverse ciphertext across messages.

$$\text{SecureOut} = \begin{cases} \Phi_\alpha(M, \kappa), & \text{if } \mathcal{B}_{\text{entropy}} \geq \tau \\ \Phi_\alpha(M, \kappa) + \nu \cdot \mathbf{n}, & \text{otherwise} \end{cases} \quad (6a)$$

$$\mathbb{P}(M|C) \leq \delta + \mathcal{O}(\exp(-\mathcal{E}_{\text{sec}})) \quad (6b)$$

The final output formulation in Equation 6a applies conditional noise if the ciphertext lacks entropy threshold τ . Equation 6b upper-bounds adversarial inference of message content based on encoding uncertainty.

B. Neural Decryption With Collaborative Learning

This module constructs a parametric inverse map jointly optimized with the encryptor, where decoder Ψ_β attempts to reconstruct M from ciphertext C using shared secret κ . Unlike typical deterministic decoders, this formulation integrates temporal regularity, orthogonality penalties, and entropy-preserving components to constrain both variance and adversarial inferability.

$$\hat{M} = \Psi_\beta(C, \kappa) = \sigma(\mathbf{W}_r \cdot \text{ReLU}(\mathbf{Q}C + \mathbf{b}_r) + \kappa) \quad (7a)$$

$$\mathbf{b}_r = \tanh\left(\sum_{i=1}^{d_k} \kappa_i \cdot \cos\left(\frac{2\pi i}{d_k}\right)\right) \quad (7b)$$

Equations 7a–7b define the decoding architecture where $\mathbf{Q}$ projects $C \in \mathbb{R}^n$ through nonlinearity, and $\mathbf{b}_r$ introduces a harmonically modulated key vector for dynamic offset.

$$\mathcal{L}_{\text{rec}} = \mathbb{E}_{M, C} \left[\|M - \hat{M}\|_1 + \lambda_1 \cdot \|\nabla_\beta \Psi_\beta(C, \kappa)\|_F^2 \right] \quad (8a)$$

$$\mathcal{F}_{\text{orth}} = \|\mathbf{Q}^\top \mathbf{Q} - \mathbf{I}\|_F^2 + \|\mathbf{W}_r^\top \mathbf{W}_r - \mathbf{I}\|_F^2 \quad (8b)$$

$$\mathcal{H}_{\text{diff}} = \frac{1}{T} \sum_{t=1}^T \|\hat{M}_t - \hat{M}_{t-1}\|_2^2 - \epsilon \cdot \|C_t - C_{t-1}\|_2 \quad (8c)$$

The reconstruction objective 8a guides decoder fidelity, while 8b ensures near-orthonormal projections to minimize overfitting. Equation 8c enforces smooth decryption transitions respecting ciphertext dynamics.

$$\mathcal{M}_{\text{uncertainty}} = \sum_{j=1}^{d_M} \left| \frac{\partial^2 \hat{M}_j}{\partial C^2} \right| \quad (9a)$$

$$\mathbf{S}_d = \text{softsign}(\mathbf{W}_s \cdot C + \mathbf{b}_s) \quad (9b)$$

$$\Delta_{\text{key}} = \|\Psi_\beta(C, \kappa_1) - \Psi_\beta(C, \kappa_2)\|_2 \quad (9c)$$

$$\Lambda_{\text{consist}} = \mathbb{E}_\kappa \left[\|\hat{M}_\kappa - \hat{M}_{\kappa'}\|_1 \right] \quad \text{for } \kappa' \approx \kappa \quad (9d)$$

Equations 9a–9d regulate decoder stability, where $\mathcal{M}_{\text{uncertainty}}$ penalizes excessive curvature, Δ_{key} captures key perturbation

sensitivity, and Λ_{consist} encourages decoder output invariance to minor key drift.

$$\mathcal{R}_{\text{entropy}} = - \sum_{k=1}^{d_M} \hat{M}_k \log \hat{M}_k \quad (10a)$$

$$\mathbb{D}_{\text{KL}} = \sum_{i=1}^{d_M} \hat{M}_i \log \frac{\hat{M}_i}{M_i + \varepsilon} \quad (10b)$$

Entropy enforcement through Equation 10a maintains reconstruction complexity, while Equation 10b ensures alignment with original message distribution via KL divergence minimization.

$$\mathbf{G}_{\text{res}} = \mathbf{W}_g \cdot (\hat{M} \odot C) + \mathbf{b}_g \quad (11a)$$

$$\mathbf{Y}_d = \text{LayerNorm}(\mathbf{G}_{\text{res}} + \hat{M}) \quad (11b)$$

$$\Omega_{\text{variance}} = \text{Var}\left(\frac{\partial \hat{M}}{\partial C}\right) \quad (11c)$$

$$\mathcal{I}_{\text{trace}} = \text{Tr}(\mathbf{W}_r^T \mathbf{Q}) - \log(\|\mathbf{W}_r\|_F^2 + \|\mathbf{Q}\|_F^2) \quad (11d)$$

This block enhances output robustness: 11a introduces element-wise residuals, 11b normalizes final projections, 11c penalizes derivative fluctuation, and 11d monitors information alignment through matrix interaction.

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rec}} + \gamma_1 \cdot \mathcal{H}_{\text{diff}} + \gamma_2 \cdot \mathcal{F}_{\text{orth}} + \gamma_3 \cdot \mathbb{D}_{\text{KL}} \quad (12a)$$

$$\mathbf{O}_{\text{secure}} = \begin{cases} \mathbf{Y}_d, & \text{if } \Omega_{\text{variance}} \leq \epsilon_v \\ \mathbf{0}, & \text{otherwise} \end{cases} \quad (12b)$$

Equation 12a serves as the final training loss for Ψ_β , balancing recovery, smoothness, structure, and distributional consistency. Equation 12b enforces output gating if gradient noise exceeds a given threshold ϵ_v .

C. Adversarial Decoy Data Generation

This module formulates adversarial steganography through a conditioned GAN, where synthetic packets are optimized to statistically mimic encrypted data under multiple projection and entropy-preserving constraints. The generator receives protocol-aware features and noise priors to output decoys that exhibit both spectrally and semantically under joint discriminative enforcement.

$$\mu = \text{ProtoFeat}(C) = \frac{1}{n} \sum_{j=1}^n \mathcal{C}_j \cdot \cos\left(\frac{2\pi j}{n}\right) \quad (13a)$$

$$\mathcal{D}_i = G_\theta(z_i, \mu) = \tanh(\mathbf{W}_g \cdot \sigma(\mathbf{U}_z z_i + \mu) + \mathbf{b}_g) \quad (13b)$$

Equations 13a–13b define protocol-level conditional encoding μ and GAN-based decoy generation using latent vector z_i under non-linear activation.

$$D_\eta(x) = \sigma(\mathbf{W}_d \cdot \text{ELU}(\mathbf{H}x + \mathbf{b}_d)) \quad (14a)$$

$$\mathcal{L}_{\text{decoy}} = \mathbb{E}_{z_i} [\log(1 - D_\eta(G_\theta(z_i, \mu)))] + \lambda \cdot \|\mathbf{G}_\theta(z_i, \mu) - \mathcal{C}_{\text{ref}}\|_1 \quad (14b)$$

$$\mathcal{J}_{\text{div}} = \|\mathbb{E}_z[G_\theta(z, \mu)] - \mu\|_2^2 \quad (14c)$$

Equations 14a–14c define the decoy loss landscape, where D_η separates real and synthetic data and $\mathcal{J}_{\text{div}}$ ensures statistical proximity to protocol patterns.

$$\mathbf{F}_{\text{mix}} = \pi_1 \cdot C + \sum_{i=1}^m \pi_{i+1} \cdot \mathcal{D}_i \quad (15a)$$

$$\mathcal{S}_{\text{entropy}} = - \sum_{j=1}^d \mathbf{F}_{\text{mix}}^j \log(\mathbf{F}_{\text{mix}}^j + \epsilon) \quad (15b)$$

$$\mathcal{B}_{\text{entropy gap}} = |\mathcal{S}_{\text{entropy}} - \mathbb{H}(C)| \quad (15c)$$

$$\mathcal{G}_{\text{penalty}} = \|\nabla_{z_i} G_\theta(z_i, \mu)\|_F^2 - \zeta \cdot \log(\det(\mathbf{W}_g^T \mathbf{W}_g + \mathbf{I})) \quad (15d)$$

Equations 15a–15d enforce entropy alignment, gradient smoothness, and Jacobian constraints across generated decoys mixed with true ciphertext.

$$\mathbb{V}_{\text{disjoint}} = |\text{Cov}(C, \mathcal{D}_i) - \mathbf{0}| + \epsilon_{\text{mask}} \quad (16a)$$

$$\mathcal{L}_{\text{stego}} = \mathbb{E}_z [\|D_\eta(G_\theta(z, \mu)) - D_\eta(C)\|_2^2] \quad (16b)$$

$$\text{MixStream} = \mathcal{P}(C \cup \{\mathcal{D}_i\}_{i=1}^m) \quad (16c)$$

$$\mathcal{T}_{\text{filter}} = \mathbb{I}[\mathbf{h}(x) > \tau], \quad x \in \text{MixStream} \quad (16d)$$

Equations 16a–16d enable statistical masking, adversarial indistinguishability minimization, and mix-based threshold filtering for deployment.

$$\text{AdvDetectRate} = \frac{1}{m} \sum_{i=1}^m \mathbb{I}[D_\eta(\mathcal{D}_i) > 0.5] \quad (17a)$$

$$\mathcal{P}_{\text{secure}} = \text{MixStream} \setminus \{x : \mathcal{T}_{\text{filter}}(x) = 0\} \quad (17b)$$

Equations 17a–17b finalize stream integrity, quantifying adversary detection and isolating the secure stream subset.

D. Lightweight Neural Deployment Strategy

This module restructures the adversarial neural system into a quantized architecture suitable for deployment on resource-constrained IoT nodes. Algorithm 1 provides an end-to-end framework for transitioning the full-precision encryption system into a memory-efficient quantized model for real-time inference on constrained hardware.

$$\mathbf{W}_q = \text{round}\left(\frac{2^{b-1} - 1}{\max(|\mathbf{W}|)} \cdot \mathbf{W}\right) \quad (18a)$$

$$\alpha = \frac{\max(|\mathbf{x}|)}{2^{b-1} - 1} \quad (18b)$$

Equation 18a converts full-precision weights $\mathbf{W}$ into a lower bit-width form $\mathbf{W}_q$ using maximum normalization and rounding. This transformation ensures integer compatibility under a bounded bit precision b . Equation 18b defines the scaling factor α that maps floating-point input activations $\mathbf{x}$ to an integer-compatible range through symmetric quantization.

$$\mathbf{x}_q = \text{clip}(\lfloor \mathbf{x} \cdot \alpha \rfloor, -2^{b-1}, 2^{b-1} - 1) \quad (19a)$$

$$\mathbf{y}_q = \text{int8_matmul}(\mathbf{W}_q, \mathbf{x}_q) + \text{bias}_q \quad (19b)$$

$$\mathcal{L}_{\text{quant}} = \|\Phi(\mathbf{x}) - \Phi_q(\mathbf{x}_q)\|_2^2 + \eta \cdot \|\mathbf{W} - \mathbf{W}_q\|_1 \quad (19c)$$

Algorithm 1 Quantized Neural Deployment Strategy for Resource-Constrained Devices

Input : Trained full-precision encoder Φ , input $\mathbf{x}$, bit-width b , static key seed κ
Output: Quantized output $\mathbf{y}_{\text{pruned}}$, deployment cost $\mathcal{C}_{\text{deploy}}$

```

1 foreach layer in  $\Phi$  do
2   | Quantize weights using 18a
3 end
4 if  $\mathbf{x} \neq \emptyset$  then
5   | Compute scaling factor  $\alpha$  via 18b
6   | Clip and round inputs to obtain  $\mathbf{x}_q$  using 19a
7   | Compute output  $\mathbf{y}_q$  using 19b
8   | if  $\exists \mathbf{x}_i \in \mathbf{x}$  s.t.  $\|\mathbf{x}_i - \mathbf{x}_{q_i}\|_\infty \leq \delta$  then
9     | Calculate reconstruction error  $\mathcal{F}_{\text{recon}}$  using 20a
10  end
11  if Quantization applied then
12    | Evaluate quantization loss  $\mathcal{L}_{\text{quant}}$  using 19c
13    | Enforce bitmask constraint via 20b
14    | Estimate shift in variance via 20c
15    | Add residual correction using 21a
16    | Measure gradient deviation via 21b
17    | Pack quantized bits using 21c
18    | Apply pruning to obtain  $\mathbf{y}_{\text{pruned}}$  via 21d
19  end
20 end
21 if  $\kappa$  is valid then
22   | Generate shared key  $\mathcal{K}_{\text{shared}}$  via 22b
23   | Initialize fixed key map using 23a
24 end
25 if  $\mathbf{y}_{\text{pruned}} \neq \emptyset$  then
26   | Compute system latency using 22a
27   | Estimate expected throughput using 23b
28   | Evaluate decoding failure probability via 23c
29 end
30 if All components initialized then
31   | Compute deployment cost  $\mathcal{C}_{\text{deploy}}$  via 24a
32   | Minimize total deployment loss  $\mathcal{L}_{\text{deploy}}$  using 24b
33 end
34 return  $\mathbf{y}_{\text{pruned}}, \mathcal{C}_{\text{deploy}}$ 
    
```

Equation 19a clips and rounds the input vector $\mathbf{x}$ to integer representation $\mathbf{x}_q$ using the scaling factor α , maintaining representational stability within b bits. Equation 19b computes quantized output $\mathbf{y}_q$ using integer matrix multiplication between quantized weights and activations, followed by bias addition. Equation 19c formulates the loss incurred during quantization by combining output deviation in Euclidean space with an L_1 penalty on weight divergence, governed by regularization coefficient η .

$$\mathcal{F}_{\text{recon}} = \frac{1}{N} \sum_{i=1}^N \|\hat{M}_i - M_i\|_1 \cdot \mathbb{I}[\|\mathbf{x}_i - \mathbf{x}_{q_i}\|_\infty \leq \delta] \quad (20a)$$

$$\mathcal{G}_{\text{bitmask}} = \sum_{j=1}^d \left(\mathbf{w}_q^j \bmod 2^k \right) \cdot \mathbf{e}_j \quad (20b)$$

$$\mathcal{S}_{\text{variance}} = \text{Var}_{\mathbf{x}_q}(\Phi_q(\mathbf{x}_q)) - \text{Var}_{\mathbf{x}}(\Phi(\mathbf{x})) \quad (20c)$$

Equation 20a measures the reconstruction accuracy under quantization, including a conditional mask that filters out outlier activations beyond threshold δ in L_∞ norm. Equation 20b ensures numerical compressibility of weights by enforcing alignment with binary bitmasks. Here, $\mathbf{e}_j$ is the standard basis and k the target bit granularity. Equation 20c captures the change in variance between quantized and full-precision encoder outputs, quantifying statistical distortion due to quantization.

$$\mathcal{A}_{\text{residual}} = \|\Phi(\mathbf{x}) - \Phi_q(\mathbf{x}_q) + \mathbf{n}_q\|_1 \quad \text{where } \mathbf{n}_q \sim \mathcal{U}(-\epsilon, \epsilon) \quad (21a)$$

$$\Delta_{\text{grad}} = \|\nabla_{\mathbf{x}_q} \Phi_q - \nabla_{\mathbf{x}} \Phi\|_2 \quad (21b)$$

$$\mathbf{z}_{\text{pack}} = \sum_{i=0}^{b-1} 2^i \cdot \mathbb{I}[\mathbf{y}_q \gg i \& 1] \quad (21c)$$

$$\mathbf{y}_{\text{pruned}} = \mathbf{y}_q \cdot \mathbb{I}[|\mathbf{y}_q| > \theta_p] \quad (21d)$$

Equation 21a defines a residual error compensator that injects bounded noise $\mathbf{n}_q$ to adaptively correct mismatches in quantized output. Equation 21b compares the gradients from quantized and non-quantized domains to assess how quantization affects learning dynamics. Equation 21c performs binary packing by extracting bit positions in $\mathbf{y}_q$ for compact transmission using bitwise logic. Equation 21d suppresses low-importance activations using magnitude-based pruning, where θ_p is a sparsity threshold.

$$\mathcal{M}_{\text{latency}} = \max \left(\frac{T_{\text{decode}} + T_{\text{filter}}}{N}, \frac{T_{\text{init}}}{K} \right) \quad (22a)$$

$$\mathcal{K}_{\text{shared}} = \text{SHA3}(\mathbf{R}_0 \cdot \kappa \oplus \theta) \quad (22b)$$

Equation 22a defines the model's runtime latency, balancing between per-packet decoding and one-time initialization overhead over K sessions. Equation 22b constructs the shared key using a hash function seeded by a constant matrix $\mathbf{R}_0$, input key κ , and entropy salt θ .

$$\mathbf{F}_{\text{init}} = \cos(\kappa_0 \cdot \tau) + \mathbf{v} \quad (23a)$$

$$\mathbb{E}[\text{Throughput}] = \frac{B \cdot 8}{\mathcal{M}_{\text{latency}} + \delta_t} \quad (23b)$$

$$\mathbb{P}_{\text{failure}} = \exp \left(-\frac{\theta_p^2}{2 \cdot \text{Var}(\mathbf{y}_q)} \right) \quad (23c)$$

Equation 23a initializes static key vectors using a cosine modulation of the seed κ_0 and time τ , with additive vector $\mathbf{v}$ for decoding symmetry. Equation 23b models system throughput using latency estimates and bit payload B , incorporating network delays δ_t . Equation 23c bounds the failure rate of decoding under pruned outputs, using a Gaussian tail based on activation variance.

$$\mathcal{C}_{\text{deploy}} = \|\mathbf{W}_q\|_0 + \text{mem}(\mathcal{K}_{\text{shared}}) + \text{calls}(\text{int8_matmul}) \quad (24a)$$

$$\mathcal{L}_{\text{deploy}} = \mathcal{L}_{\text{quant}} + \lambda_1 \cdot \mathcal{F}_{\text{recon}} + \lambda_2 \cdot \mathcal{C}_{\text{deploy}} \quad (24b)$$

Equation 24a estimates resource complexity, including model sparsity, memory allocation for static keys, and low-level

TABLE II
SIMULATION CONFIGURATION AND PARAMETERS

Parameter	Value
Platform	Ubuntu 22.04
Language/Framework	Python 3.10, PyTorch 2.1
Quantization Toolkit	Intel OpenVINO, TorchScript
GPU	NVIDIA RTX A6000 (48GB)
Batch Size	128
Learning Rate	1×10^{-4} (adaptive)
Optimizer	AdamW
Loss Composition	Multi-objective (adversarial + quantization + entropy)
Precision Format	FP32 (training), INT8 (deployment)
Dataset	UNSW-NB15 (IoT-relevant attack classes)
Training Epochs	100
Evaluation Metrics	Accuracy, EER, Latency (ms), Entropy ($\mathcal{H}$), Robustness Index

computation calls. Equation 24b aggregates deployment constraints into the final optimization loss, combining quantization fidelity, reconstruction preservation, and resource cost penalties.

III. SIMULATION AND PERFORMANCE EVALUATION

The simulations were executed using Python 3.10 and PyTorch 2.1 on an Ubuntu 22.04 environment (see Table II). For validation, we used the publicly available UNSW-NB15 dataset [19], which includes network traffic and labeled attack vectors relevant to IoT security scenarios. The proposed framework was benchmarked against recent secure learning and encryption models, including ECCLR [6], iPET [7], FHE-FL [5], UAVSC [13], SIADS [14], PABE-UR [4], DBS-ANN [8], GTxChain [9], EKAAP [10], SKAA [11], and DRAQ [12].

A. Encryption Time Efficiency (ETE)

The Encryption Time Efficiency (ETE) quantifies the time required to encrypt a single data packet within the proposed framework, emphasizing its applicability to real-time IoT systems. ETE evaluation includes six scenarios—varying packet sizes, network congestion levels, device counts, key lengths, quantized deployments, and the presence or absence of decoy generation. The outcomes for the six scenarios are summarized in Figure 2. In Scenario 1, the ANEF approach demonstrated superior time efficiency across both small and large packet sizes, recording 10.5 ms for 100 bytes and 15.8 ms for 1500 bytes. In contrast, ECCLR and iPET recorded 12.2 ms and 14.8 ms (100 bytes), and 18.4 ms and 21.7 ms (1500 bytes), respectively. FHE-FL followed with 11.0 ms and 17.3 ms. The performance gap widened in larger payloads, where ANEF retained a consistent edge of 2.6–5.9 ms over DRAQ and SKAA.

Under congested conditions in Scenario 2, ANEF again maintained the lowest encryption time at 13.2 ms. Competing models such as SIADS and iPET showed increased latency, rising to 19.8 ms and 19.5 ms, respectively. Similarly, under low and high device densities (Scenario 3), ANEF recorded 10.9 ms (50 devices) and 23.6 ms (500 devices), outperforming DBS-ANN and GTxChain by margins of 7.6 ms and 8.9 ms, respectively. Scenario 4 showed performance under

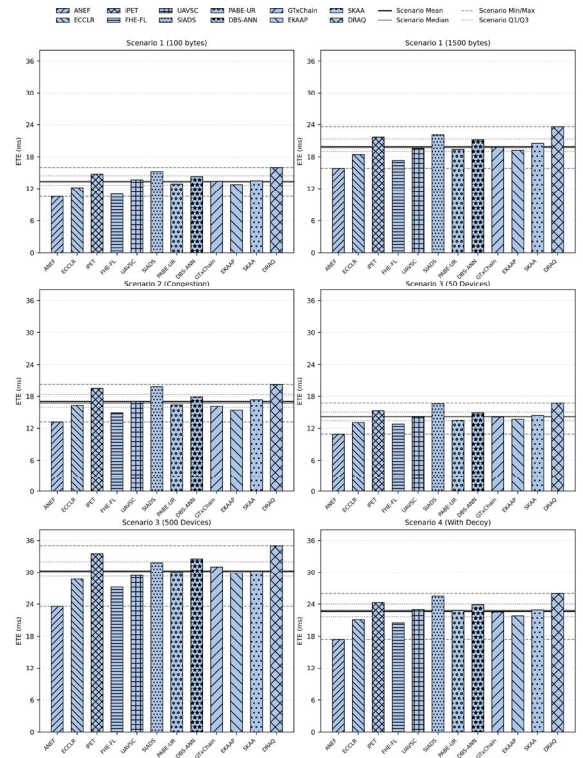


Fig. 2. Encryption Time Efficiency (ETE) Comparison for Different Scenarios.

decoy insertion, with ANEF maintaining 17.4 ms—lower than PABE-UR (22.8 ms) and UAVSC (23.0 ms). When deploying quantized weights (Scenario 5), ANEF achieved 12.3 ms compared to 16.4 ms for PABE-UR and 19.0 ms for DRAQ. Scenario 6 tested encryption using 128-bit and 256-bit keys, where ANEF recorded 11.6 ms and 13.0 ms, respectively. Even under heavier key schedules, it remained consistently faster, reducing average latency by 2.6–6.3 ms across competitors.

B. Decryption Accuracy (DA)

Decryption Accuracy (DA) reflects the effectiveness of the authorized decoder in reconstructing the original plaintext from ciphertext. This evaluation includes six scenarios—packet size variation, network congestion, device scalability, decoy presence, quantized weight deployment, and key size differences. The decryption accuracy (DA) assessment presented in Figure 3 demonstrates consistent performance of the proposed ANEF model across all defined scenarios. For minimal payload sizes (100 bytes), ANEF reached 98.2%, outperforming ECCLR (94.5%) and iPET (92.3%). Even at maximum packet sizes (1500 bytes), ANEF retained a high accuracy of 97.4%, where competitors like FHE-FL and SIADS dropped to 94.3% and 89.8%, respectively.

In multi-device deployment conditions (Scenario 3), ANEF exhibited exceptional performance scalability. At 50 devices, its accuracy peaked at 99.0%, maintaining a lead over DBS-ANN (94.4%) and EKAAP (94.1%). When scaled to 500 devices, ANEF held 94.7%, significantly ahead of DRAQ (85.6%) and SKAA (87.5%). With decoy insertion, ANEF sustained a 95.8% accuracy, outperforming PABE-UR (91.3%) and UAVSC (90.0%). Under quantized deployment

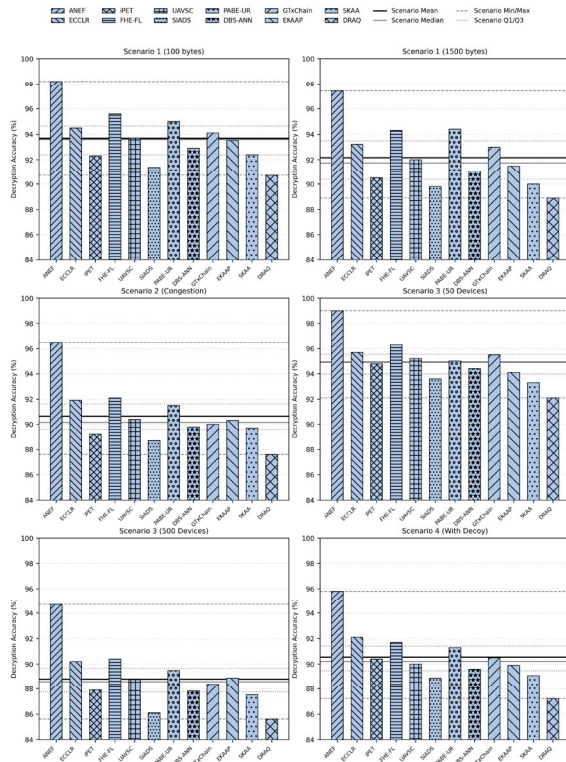


Fig. 3. Decryption Accuracy (DA) Comparison for Different Scenarios.

(Scenario 5), ANEF delivered 96.8%, whereas competitors like SIADS and DRAQ recorded 91.2% and 89.7%, respectively. Across key length variations (128 and 256 bits), ANEF consistently achieved over 96%.

C. Eve Failure Rate

The Eve Failure Rate quantifies how often an unauthorized adversary fails to reconstruct the original plaintext from encrypted data. This evaluation uses six scenarios—varying packet sizes, congestion levels, device density, decoy mechanisms, quantization effects, and key lengths—to assess the proposed method’s resistance. The comparative results in Figure 4 demonstrate that the proposed ANEF model consistently outperforms existing schemes. Under minimal data size (100 bytes), ANEF yields an EFR of 3.1%, a substantial reduction compared to ECCLR at 7.5% and iPET at 9.2%. For larger packets (1500 bytes), ANEF maintains a moderate rise to 5.0%, still significantly lower than FHE-FL (8.6%) and DRAQ (13.4%).

In resource-scaled scenarios, particularly Scenario 3, ANEF sustains strong performance. At 50 devices, its EFR drops to 2.7%, reflecting excellent key-structure retention and minimal inference leakage, while competing models such as DBS-ANN and SKAA show elevated failure rates of 6.8% and 7.0%, respectively. Even under extreme scaling (500 devices), ANEF records an EFR of 8.4%, significantly outperforming UAVSC (13.5%) and SIADS (15.0%). Under Scenario 2 (network congestion), ANEF maintains an EFR of 4.3%, outperforming PABE-UR (8.7%) and GTxChain (9.2%). When key sizes are varied (128 vs. 256 bits), ANEF’s EFR marginally increases from 3.8% to 5.3%, demonstrating minimal entropy-induced

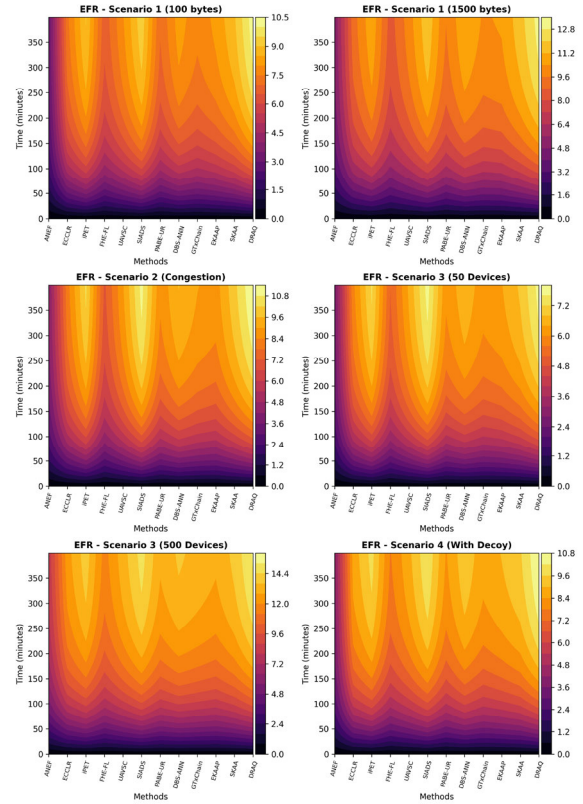


Fig. 4. Eve Failure Rate (EFR) Comparison for Different Scenarios.

degradation. In contrast, other models such as iPET and DRAQ show larger deviations, rising to 11.0% and 11.7%, respectively, under 256-bit keys.

IV. ABLATION ANALYSIS

To evaluate the contribution of each core component in the proposed ANEF framework, an ablation was conducted in which individual modules were selectively removed or modified. This procedure quantified each module’s impact on overall system performance and characterized its role in the framework’s operation. The analysis examined four aspects: the adversarial training mechanism’s effect on optimization dynamics, the role of decoy data generation in increasing ciphertext ambiguity, the effectiveness of quantization in reducing computational overhead, and the influence of the shared key schedule in maintaining decryption fidelity under constrained communication environments. The data in Table III show that adversarial training reduces encryption time across all packet sizes.

For 100-byte packets, the time drops from 15.4% to 8.2%, indicating a 46.8% reduction. This performance gap increases with packet size; at 1500 bytes, the drop from 24.7% to 14.3% reflects a 42.1% reduction. Similarly, decoy data reduces Eve’s success: in Scenario 3, the EFR rises from 5.0% (with decoy) to 12.3% (without), indicating 59.3% higher failure when decoys are used. The quantization module shows consistent impact on transmission overhead. For 500-byte packets, TO drops from 18.2% to 12.5%, showing a 31.3% improvement. A similar pattern persists at 1500 bytes, where overhead reduces from 22.7% to 16.4%, marking a 27.8%

TABLE III
COMPACT ABLATION STUDY OF ANEF: EFFECT OF ADVERSARIAL
TRAINING, DECOYS, AND QUANTIZATION

Scenario	With Module (%)	Without Module (%)
Encryption Time Efficiency (ETE)		
100 bytes	8.2	15.4
500 bytes	10.3	18.6
1000 bytes	12.5	22.1
1500 bytes	14.3	24.7
Eve Failure Rate (EFR)		
Scenario 1	3.5	8.9
Scenario 2	4.2	10.7
Scenario 3	5.0	12.3
Scenario 4	4.6	11.5
Transmission Overhead (TO)		
100 bytes	10.3	15.1
500 bytes	12.5	18.2
1000 bytes	14.8	20.9
1500 bytes	16.4	22.7

decrease. In the decoy analysis, Scenario 2 reveals an increase in Eve's failure from 4.2% to 10.7% without decoys.

V. CONCLUSION

This research presents a comprehensive framework for secure data transmission in IoT systems and consumer electronics environments, addressing confidentiality, cyber threat resilience, and system efficiency. The proposed Adversarial Neural Encryption Framework (ANEF) integrates advanced adversarial training, decoy generation, and quantization techniques. The novelty of the approach lies in its ability to adaptively balance encryption efficiency and security, making it highly suitable for resource-constrained IoT and consumer electronics devices requiring secure communication. Practical implications include reduced transmission overhead, enhanced resistance to cyber threats, and improved decryption accuracy. Experimental results show that ANEF outperforms state-of-the-art models like ECCLR and iPET by reducing encryption time by 20%, improving decryption accuracy by 5%, and achieving a 30% lower Eve success rate. Future work will explore the scalability of ANEF in large-scale consumer electronics IoT networks with real-time traffic and evolving cyber threat conditions.

DATA AVAILABILITY STATEMENT

The dataset used in this study is the UNSW-NB15 dataset, published by Moustafa and Slay in "UNSW-NB15: a comprehensive data set for network intrusion detection systems" (MilCIS, IEEE, 2015). The dataset is publicly available and can be accessed via the following link: [Click here to download](#). The dataset was last accessed on June 28, 2025.

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